

NGUYEN GIA HUY

DevOps Engineer

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PROFESSIONAL SUMMARY

DevOps engineer with nearly three years of experience building and operating infrastructure for fintech, banking, and utility platforms. At VNPAY Cloud, designed and operated an OpenStack private cloud running **50+ production Kubernetes clusters**, automated the cluster lifecycle with Cluster API (provisioning, scaling, and zero-downtime upgrades), and automated about **80%** of data center provisioning. At Vikki Digital Bank, deployed Thought Machine Vault Core on AWS EKS with its CI/CD pipeline and made the simulation test stage **20x faster**. Manages infrastructure as code with Terraform and AWS SAM and monitors production with Prometheus, Grafana, CloudWatch, and DataDog. Completed the DevOps on AWS Specialization (Coursera).

TECHNICAL SKILLS

Container Orchestration: Kubernetes, Cluster API, Amazon EKS, Amazon ECS, Docker
Cloud Platforms: AWS (EKS, ECS, ECR, ELB, Lambda, RDS, S3, CloudWatch), OpenStack
Infrastructure as Code: Terraform, AWS SAM
CI/CD & Release: CI/CD pipeline design (build, test, deploy), Jenkins, ArgoCD, Git, Amazon ECR
Observability: Prometheus, Grafana, OpenTelemetry, ELK, CloudWatch, DataDog
Access & Traffic Control: Kong Gateway, Keycloak (RBAC, multi-tenancy), HashiCorp Vault, Consul
Automation & Tooling: Golang, Python

WORK EXPERIENCE

Middle Software Engineer – Cloud Infrastructure & Automation 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Deployed a new **Thought Machine Vault Core** version to AWS, standing up the infrastructure that the bank's core banking platform runs on.
- Provisioned and integrated **Amazon EKS** for the platform's workloads, **RDS** for its database, **ECR** for container images, and **Elastic Load Balancing** for traffic.
- Set up continuous integration and delivery for Vault Core: a **CI/CD pipeline** that automates the build, test, and deployment stages.
- Rebuilt the simulation testing framework in **Golang**, replacing the Gauge-based framework and cutting test runtime from **24 minutes to just over 50 seconds (20x faster)**.
- Deployed the Golang account migration system as **ECS** tasks and **Lambda** functions over RDS; migration runs dropped from **2.5 hours to 27 minutes (5x)**.
- Connected the deployed platform to the bank's external backend services for core banking operations and transaction workflows.
- Also delivered application work on the platform: Lending and Overdraft products in Python Smart Contracts (interest accrual, penalty and late fees, repayment, settlement).

Software Engineer – Data Platform Infrastructure 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Built and maintained the data platform's AWS infrastructure as code with **Terraform**.
- Deployed the serverless pipeline components with **AWS SAM**, alongside containerized workloads on Amazon ECS.
- Set up **CloudWatch** monitoring for pipeline health and failures across production workloads.
- Added **DataDog** monitoring for operational anomaly detection on a platform serving millions of users.
- Kept production ETL running across Lambda, ECS, RDS, S3, and Athena, the workloads that feed au Denki's analytics and operational systems.
- Lifted batch throughput by **150%** by parallelizing execution, redesigning pipelines, and making data access leaner.
- Carried the search platform through its move from **Elasticsearch to OpenSearch**, with redesigned ingestion and indexing flows for stability, consistency, and query performance.

Cloud Platform Engineer (Software Engineer) 04/2024 – 04/2025
VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

Private Cloud and Cluster Lifecycle

- Designed and operated the **OpenStack** private cloud behind VNPAY Cloud, the platform serving fintech and enterprise customers.
- Operated **50+ production Kubernetes clusters** on that platform across their full lifecycle.
- Automated cluster **provisioning** with **Cluster API**, managing clusters declaratively through Kubernetes-style resources.

- Automated cluster **scaling** through the same Cluster API-based lifecycle, adjusting capacity without hand-built infrastructure changes.
- Delivered **zero-downtime upgrades** on production clusters by rolling changes through cluster machines without interrupting running workloads.

Infrastructure Automation

- Automated about **80%** of data center provisioning, turning manual setup work into repeatable automation.
- Automated infrastructure configuration processes across the data center, significantly reducing manual operational effort.

Traffic, Identity, and Multi-tenancy

- Integrated **Kong Gateway** into a microservices ecosystem of **100+ services** and operated it in production.
- Implemented service-level and network-level access controls in Kong, governing access to each service at both layers.
- Implemented multi-tenant authentication with **Keycloak**, giving each of **20+ customer organizations** its own isolated identities.
- Implemented role-based access control on top of Keycloak to enforce tenant isolation on shared infrastructure.

Blockchain Developer

06/2023 – 09/2023

EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Built event-driven services that capture and process on-chain events and move them into off-chain databases and APIs.
- Connected them to **5+ networks** (Ethereum, BSC, Polygon, Aptos, Sui) with consistent transaction handling and data integrity.
- Wrote and reviewed **15+** DeFi and NFT smart contracts (Solidity, Move).

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Implemented **multi-cluster failover** for the backend, keeping the platform at **99.95% uptime**.
- Built the observability stack: OpenTelemetry instrumentation, Prometheus and Grafana metrics, ELK logging, and ClickHouse.
- Built a gRPC-Gateway, Kafka, MongoDB, and Redis backend serving up to **1.5K requests/second** at **80–100 ms** latency.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting platform on LiveKit with OpenAI-powered live transcription, voice translation, and summarization, including its real-time translation pipeline.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera